



HT-ProMon Series Industrial Digital Heat Trace Controller

User Manual

Version 2.0

Table of Contents

Table of Contents.....	1
1. Product Overview & Introduction.....	5
1.1 Purpose of This Manual.....	5
1.2 Product Overview.....	5
1.3 Models Covered.....	6
2. Safety Precautions.....	7
WARNING — Electrical Shock Hazard.....	7
WARNING — Ground Fault Hazard.....	7
WARNING — Unintended Heater Energization.....	7
CAUTION — Stuck Relay / Welded Contacts.....	8
CAUTION — RTD Sensor Fault Safe-State.....	8
NOTICE — Configuration Persistence After Power Loss.....	8
3. Product Description.....	9
3.1 Front Panel / Display.....	9
Integrated 4-inch Touchscreen HMI (standard).....	9
3.2 Status Indicators.....	10
3.2.1 Status LED Behavior.....	10
3.2.2 RTD Status LED Indicators.....	11
3.2.3 RTD Fault Behavior.....	11
3.3 Factory Reset.....	12
4. Energizing the Controller.....	13
4.1 Pre-Energization Checklist.....	13
4.2 Initial Power-Up Sequence.....	13
4.3 First-Time Configuration.....	14
4.3 First-Time Configuration (Continued).....	15
5. Configuration.....	16
5.1 Accessing Configuration.....	16
5.1.1 Display Settings.....	16
5.1.2 Security and PIN Lock.....	17
5.1.3 Setting a PIN for the first time:.....	18
5.2 Temperature Setpoints.....	20

5.2.1 Implementing More than 1 RTD	22
5.2.2 Multi-RTD Control Modes	23
5.2.3 Renaming RTD Sensors	23
5.2.4 Zone Renaming	23
5.3 Temperature Alarms	24
5.3.1 Types of Temperature Alarms	24
5.3.2 RTD Threshold Temperature Alarms	24
5.3.3 Control Temperature Alarms	25
5.4 Electrical Settings	26
5.4.1 Ground Fault Current Protection	27
5.4.2 Load Current Protection	28
5.4.3 Stuck Relay Detection	30
5.4.4 Voltage Alarms	31
5.6 Output Operating Modes	33
5.6.1 EMR Mode (Electromechanical Relay)	33
5.6.2 SSR Mode (Solid State Relay with Time-Proportional Output)	34
5.7 Output Configuration (SSR Models)	34
5.7.1 PID Parameters (SSR Mode)	35
5.7.3 Current Ceiling The current ceiling limits the maximum heater duty cycle to keep the estimated load current within a configured limit, regardless of what the PID algorithm demands.	36
5.7.2 SSR Application Output	37
5.7.4 Auto-Tune (SSR Mode)	38
6 Alarm Relay Behavior	39
6.1. Alarms vs. Trips	39
6.1.2 Alarm Types	40
6.1.3 What Happens When an Alarm Activates	40
6.2 Trip Types	41
6.2.1 What Happens When a Trip Activates	41
6.2.3 Resetting Alarms and Trips	41
6.2.4 Critical Alert Overlay (Dashboard)	42
6.2.5 Latched Events Counter	42
7. RS-485 / Modbus RTU Reference	43
7.1 Port Configuration	43
7.2 Protocol Summary	43

7.3 Register Map.....	43
7.3.1 Input Registers (30001–30038)	43
7.3.2 Holding Registers (40001–40192)	45
Group 1 — Control Settings (40001–40008).....	45
Group 2 — Protection Thresholds (40009–40015)	46
Group 3 — Alarm Management (40016–40033)	46
Group 4 — Advanced Sensing (40034–40040).....	47
Group 5: System, Diagnostics, and Self-Test.....	47
Appended Output-Control Block: 40068–40079.....	48
PID Auto-Tune Block (40080–40082).....	49
Read-Only LoRa Block (40083–40094).....	49
Device Options, Cutout Status, and Version (40095–40122).....	50
Name String Registers (40125–40192)	51
7.3.3 Coils (00001–00038)	52
7.4 Reading Data.....	53
Example 1: Read all three RTD temperatures (FC04)	53
Example 2: Read PID gains (FC03).....	54
Example 3: Read alarm and trip status coils (FC01)	54
7.5 Writing Configuration.....	54
Example 1: Set the temperature setpoint (FC06).....	54
Example 2: Write multiple PID gains in one transaction (FC16)	55
Example 3: Clear latched alarms (FC05)	55
Write Validation and Cautions	55
8. Troubleshooting	57
8.1 Symptom and Corrective Action Reference.....	57
8.2 PT100 Resistance Reference Table.....	59
9. Maintenance	59
9.1 Routine Inspection	59
9.2 Fuse Replacement.....	60
10. Specifications.....	61
Electrical Specifications.....	61
Output and Relay Specifications.....	61
Wiring and Installation	62

Environmental Specifications.....	62
Physical Specifications	62
11. RTD Measurement Accuracy and Resolution.....	63
12. Warranty and Support	63

1. Product Overview & Introduction

1.1 Purpose of This Manual

This manual covers operation and configuration of the HT-ProMon Series Digital Heat Trace Control. It describes the controller's status indicators and interfaces, explains how to configure temperature setpoints and alarm thresholds, and serves as the reference for normal operating behavior and troubleshooting.

1.2 Product Overview

The HT-ProMon Series digital heat trace control combines temperature control with built-in power metering, ground fault monitoring, and multiple communication interfaces, making it suitable for demanding industrial heat trace applications including freeze protection and process heating. Using up to three PT100 RTD sensors per controller, users can configure zones for a variety of process and pipe temperature monitoring conditions. All controller settings—temperature setpoints, alarms, electrical protection, and remote communication can be easily configured using the integrated touchscreen HMI.

Key capabilities:

Capability	Description
Three PT100 RTD temperature sensing inputs	Up to three independent temperature sensors; any combination can be used for control or monitoring
True RMS power metering with ground fault detection	Continuous measurement of load current and ground fault leakage current; ground fault detection in less than 20 ms
Dual RS-485 Modbus RTU ports	Two Modbus RTU ports, with port 1 as the default for HMI, share a register map and support SCADA/DCS integration.
Integrated 4"×4" high-resolution HMI (720p)	Dedicated operator panel providing real-time monitoring and full configuration access without opening the enclosure.

1.3 Models Covered

This Manual covers the HT-ProMon-EMR (Electromechanical output 30A Max output) and the HT-ProMon-SSR; (Solid state relay) with integral heat sink capable of using time-proportioning (duty cycle) control; maximum load current per name plate.

Model	Description
HT-ProMon-EMR-PMB	Electromechanical relay output with pipe mounting bracket kit.
HT-ProMon-EMR-WM	Electromechanical relay output with wall mounting kit.
HT-ProMon-SSR-PMB	Solid state relay with integral heat sink; time-proportioning (duty cycle) control; maximum load current per name plate. Includes pipe mounting kit.
HT-ProMon-SSR-WM	Solid state relay output with integral heat sink; time-proportioning (duty cycle) control; maximum load current per name plate. Heatsink Machined for wall mounting applications.

NOTE

Wiring, mounting, and enclosure installation are covered in the HT-ProMon Series Installation Instructions (supplied separately). This manual covers configuration and operation only.

2. Safety Precautions

For installation-phase safety precautions — including hazards associated with mounting, conduit entry, and wiring — refer to the **HT-ProMon Series Installation Instructions**.

The following precautions apply during powered operation and configuration of the HT-ProMon controller.

WARNING

WARNING — Electrical Shock Hazard

The heater output circuit operates at line voltage. Energized conductors at the heater output terminals present a risk of electric shock.

Contact with energized heater circuit conductors can cause death or serious injury.

De-energize and lock out all power — including both control power and the heater supply circuit — before servicing heating cables, RTD sensors, or output terminals. Do not assume the heater output is de-energized because the controller indicates the output is OFF. Verify the absence of voltage with a calibrated meter before touching any conductor.

WARNING

WARNING — Ground Fault Hazard

The controller monitors ground fault leakage current in the heater circuit. A ground fault condition indicates line-to-ground leakage, typically caused by insulation damage or moisture ingress in the heating cable.

Even milliamp-level ground fault current through the body can cause lethal ventricular fibrillation.

Do not reset a ground fault trip by simply cycling power or issuing a reset command. Before resetting, physically inspect the heating cable and all connections for insulation damage, moisture, or corrosion. Do not return the controller to service with a persistent ground fault present.

WARNING

WARNING — Unintended Heater Energization

Configuration changes made via Modbus take effect immediately. Lowering the temperature setpoint below the current process temperature, or re-enabling a previously tripped output, will energize the heater circuit without warning.

Unexpected heater energization during maintenance activities can cause burns, fire, or equipment damage.

Confirm that the heater circuit is in a safe state before making any configuration changes that lower the setpoint or re-enable a tripped output.

When working on a heat trace circuit, de-energize and lock out the heater supply independently of any controller setting.

⚠ CAUTION

CAUTION — Stuck Relay / Welded Contacts

The controller continuously monitors for current flow when the output is commanded off. A welded relay contact or failed SSR can cause the heater to remain energized regardless of the controller output command.

An undetected welded contact will result in continuous heater energization, which can cause pipe overtemperature, insulation damage, or fire.

Do not acknowledge and clear a Stuck Relay trip without first inspecting the relay or SSR for contact welding or failure. Replace the relay or SSR as required before returning the controller to service.

⚠ CAUTION

CAUTION — RTD Sensor Fault Safe-State

If a PT100 sensor assigned to the control function faults or is disconnected, the controller immediately de-energizes the heater output to prevent uncontrolled temperature rise. This is the intended safe-state behavior.

In freeze-protection applications, heater shutdown on RTD fault can result in pipe freezing and associated equipment damage.

Monitor RTD sensor health via Modbus. Enable the RTD failure alarm to receive immediate notification of a sensor fault. For critical freeze-protection circuits, install redundant PT100 sensors and configure the controller to use multiple sensors in a safe control mode.

⚠ NOTICE

NOTICE — Configuration Persistence After Power Loss

All configuration parameters — including temperature setpoints, alarm thresholds, and enable/disable settings — are stored in non-volatile memory and are retained across power cycles and firmware updates.

Trip states are cleared on power restoration; the controller restarts in a non-tripped, heater-off state.

Verify all setpoints and thresholds after any firmware update to confirm that configuration has been retained as expected.

3. Product Description

3.1 Front Panel / Display

The HT-ProMon Series controller is a two-part control system consisting of the HT-ProMon touchscreen HMI and an HT-ProMon control board (EMR or SSR model). Together, the HMI and control board provides a streamlined system for operator configuration, monitoring, and heater output control.

Integrated 4-inch Touchscreen HMI (standard)

The integrated 4-inch touchscreen HMI is the primary method for setting setpoints, viewing status, and performing general configuration. The HMI communicates with the base board over the Modbus RS-485 link. The Home Screen provides a live view of key values (temperatures, load current, ground fault current, output state, and alarms) without opening the enclosure.

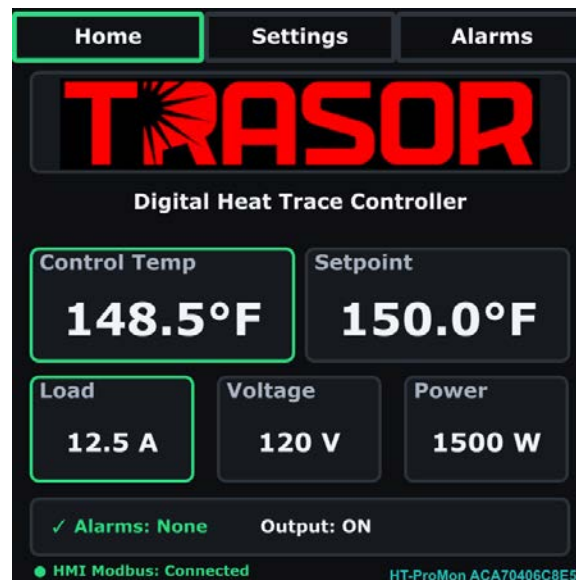


Figure 3.1 - System Home Screen

The HMI interface includes three persistent navigation (**Home**, **Settings**, and **Alarms**). Settings provides access to control, protection, and communications parameters; Alarms lists displays active and latched alarm conditions.

NOTE: The base board can continue to control the heat trace circuit using the last saved configuration if the HMI is unavailable; however, operation without the HMI is not the intended normal mode. In this condition, status is limited to the four RGB LED indicators on the base board, and configuration changes should be performed via Modbus RTU (RS-485).

3.2 Status Indicators

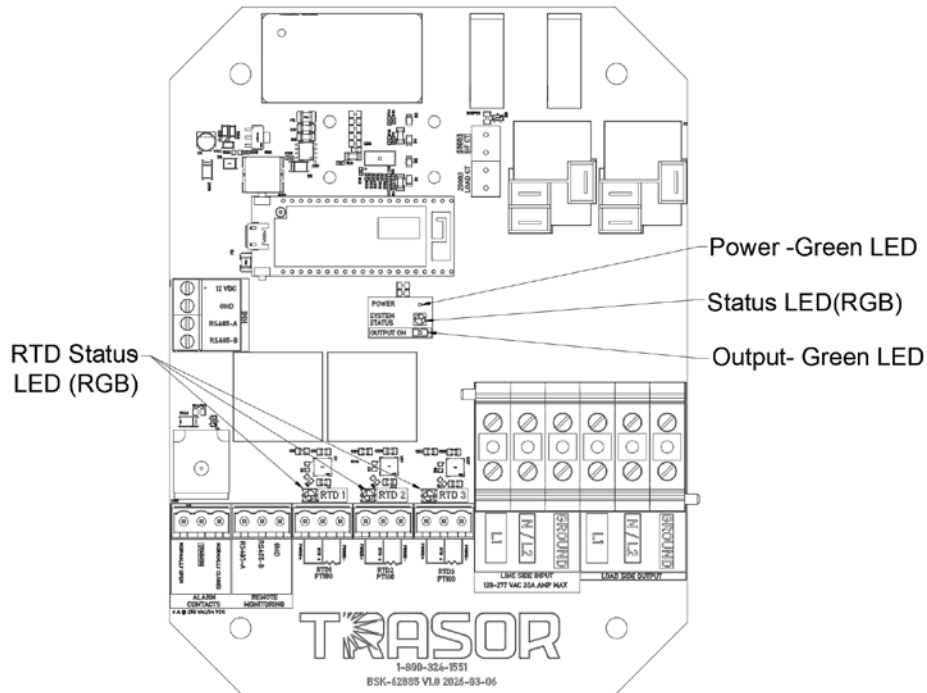


Figure 3.2 - HT-ProMon-EMR Control Board

The HT-ProMon controller has two green LEDs and four RGB LED indicators

- A green “Power On” LED indicating the board has power
- A green “Output On” LED indicating when the output state is active.
- Single Status RGB LED provides overall system and communication status.
- Three Individual RTD Status RGB LEDs correspond to RTD sensors 1–3 and indicate each sensor’s health when that sensor is configured for control or monitoring.

3.2.1 Status LED Behavior

Indicator	Color / Pattern	Meaning
System Status	Slow green blink	Boot/initialization period (first ~5 seconds)
	Solid green	Normal operation (no alarms, no trips)
	Solid amber	Alarm active or latched (no trip)
	Solid red	High-severity trip latched (heater output locked off)

3.2.2 RTD Status LED Indicators

- *Solid Red: RTD is faulted.*
- *Solid Green: RTD resistance is within the temperature converter's valid detection range.*

NOTE: RTD Status LEDs (1–3) illuminate only when that sensor has been assigned to a Control or Monitoring role. LEDs for sensors that are not configured remain off.

3.2.3 RTD Fault Behavior

If a control RTD sensor faults (invalid reading), the controller immediately de-energizes the heater output to prevent uncontrolled operation. The RTD Failure alarm is enabled by default and is latching. (Settings > Control Settings > Control Alarms)

If the alarm is not configured to latch the heater resumes automatically once the sensor recovers — no trip state or manual reset is required for RTD faults (as distinct from current trips, which always require manual reset).

NOTE: In freeze-protection applications, if the heater's output is shutdown on RTD fault can result in pipe freeze. Monitor RTD sensor health via the Dashboard or Modbus. Install redundant RTD sensors on critical circuits and configure the control mask to use multiple sensors.

3.3 Factory Reset

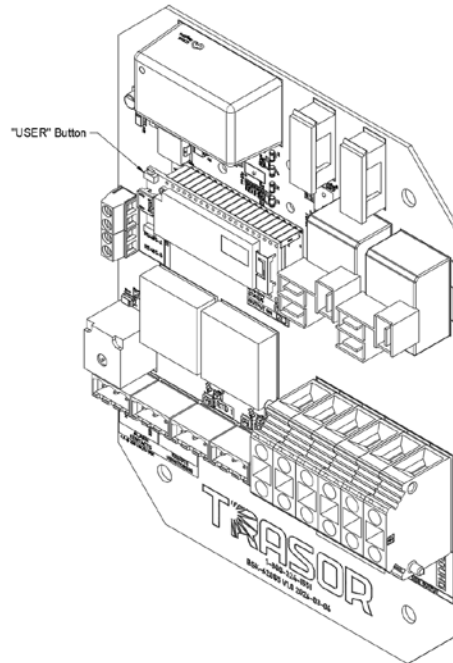


Figure 3.3 - "USER" Button Location

The HT-ProMon-EMR and HT-ProMon SSR Control Boards have one user-accessible hardware control: a small push-button on the HT-ProMon microcontroller nearest the USB-C serial port labeled "USER". The button must be pressed and held for 10 seconds to initiate a factory reset.

Factory Reset erases all settings including setpoints, thresholds, and communication parameters. Use Factory Reset only when directed by Trasor technical support or when decommissioning a unit for redeployment.

The factory reset sequence is indicated by all four LEDs cycling through blue, amber, and magenta. On successful completion, all four LEDs display solid green for two seconds, then the controller reboots. If the reset fails, all four LEDs display solid red for two seconds.

4. Energizing the Controller

4.1 Pre-Energization Checklist

Before applying power, verify that all installation and wiring steps in the *HT-ProMon Series Installation Instructions* are complete. Do not energize the controller until the installation instructions' final pre-energization checklist has been fully satisfied.

4.2 Initial Power-Up Sequence

When supply power is applied, the HT-ProMon controller performs a brief initialization sequence. The following events are visible:

1. **Status LEDs initialize.** The four RGB status indicators cycle through colors briefly as the controller starts. Normal illumination follows: LED 0 (System Status) blinks green slowly during the first 5 seconds, then transitions to solid green once the boot period ends.
2. **RTD Status 1, 2, and 3 illuminate.** Each of these corresponds to one RTD sensor. A sensor that is configured and reading correctly shows solid green. A sensor that is configured but faulted shows solid red. A sensor that is not configured for the current mode is dark (off). This gives an immediate visual summary of RTD health at power-up.
3. **Temperature monitoring begins.** All configured RTD sensors begin reporting readings. The controller starts evaluating temperatures against the configured setpoint.
4. **Heater control becomes active.** The controller begins making heater on/off decisions according to the factory default setpoint. The heater output is initialized off at boot and will energize if the measured control temperature is below the configured switching threshold.
5. **Modbus RTU interfaces start.** Both RS-485 ports begin accepting Modbus RTU commands within a few seconds of power-up.
7. **Stuck relay monitoring activates.** A 5-second startup grace period elapses before stuck relay detection becomes active. This prevents false trips from relay contact settling at power-up.

If an alarm or trip condition exists at power-up (for example, a configuration that would immediately trigger a trip), the appropriate LED states are shown and the alarm relay responds according to its configured behavior.

NOTE: The HMI (if installed) displays a splash screen for approximately 3 seconds after its own power-up, then navigates to the Home screen and begins polling the controller automatically.

4.3 First-Time Configuration

The controller ships with the following factory default settings. Review and adjust these settings for your application before placing the unit in service.

Parameter	Default Value
Control Setpoint	68.0 °F
Hysteresis (deadband)	4.0 °F
Temperature Units	°F
RTD Control Mask	RTD 1 only
RTD Monitor Mask	None (disabled)
Multi-RTD Mode	Low (lowest reading)
RTD High Limit	450.0 °F
Load Current — Low Alarm	1.0 A
Load Current — High Alarm	29.0 A
Load Current — Trip	30.0 A
Ground Fault — Alarm	20 mA
Ground Fault — Trip	30 mA
Stuck Relay Min Current	0.3 A
Stuck Relay Delay	500 ms
Alarm Relay — Failsafe Mode	Disabled
Modbus Instance 1 — Slave ID	1
Modbus Instance 1 — Baud Rate	115200
Modbus Instance 2 — Slave ID	1
Modbus Instance 2 — Baud Rate	9600

4.3 First-Time Configuration (Continued)

Steps required before placing the unit in service:

1. Set the control setpoint to the required temperature for the application (see Section 5.2).
2. Set the hysteresis appropriate to the process (see Section 5.2).
3. Verify that the RTD control configuration matches the physically installed sensor(s) (see Section 5.2.1).
4. Set load current alarm and trip thresholds for the circuit rating (see Section 5.4.2).
5. Set ground fault thresholds appropriate for the installation (see Section 5.4.1).
6. If remotely Monitoring Adjust Modbus slave IDs if multiple devices share the same RS-485 bus (see Section 7).

5. Configuration

5.1 Accessing Configuration

The HT-ProMon controller provides 2 independent interfaces for viewing and changing configuration. All interfaces read and write the same configuration state — a change made through one interface is immediately reflected through all others.

Interface	Connection	Primary Use Case
HMI	Modbus RTU over RS-485 (integral wiring)	Primary operator interface. Full configuration access through labeled menu screens.
RS-485 / Modbus RTU	2-wire RS-485; two independent ports	SCADA / DCS / HMI integration. Read and write holding registers directly.

Using the Dashboard: Navigate using the top navigation bar — **Home**, **Settings**, and **Alarms**. Settings sub-screens are organized by function. All tappable fields display a numeric keypad or selection list. Changes take effect immediately and are written to the controller over the Modbus link.

Using RS-485 / Modbus RTU: Send standard Modbus RTU function codes to the configured slave ID. The full register map is in Section 7.

5.1.1 Display Settings

Display behavior is configured under **Settings** → **Display Settings** on the Dashboard.

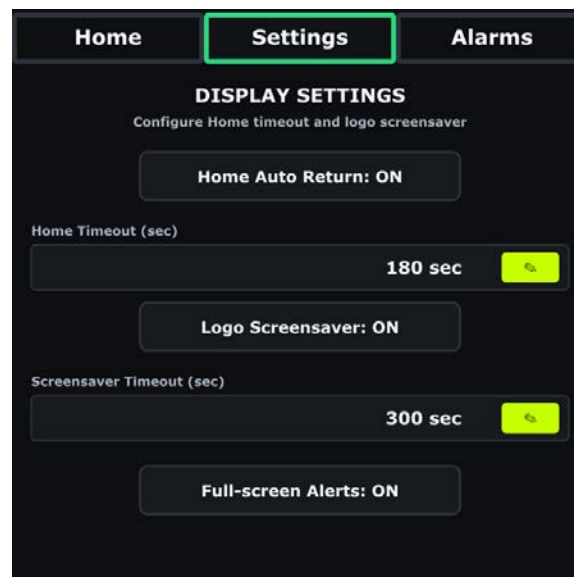


Figure 5.1.1 - Display Setting Screen

Parameter	Default	Options	Notes
Home Auto Return	ON	ON / OFF	Returns display to Home screen after timeout
Home Timeout	180 seconds	Configurable	Inactivity period before auto-return
Logo Screensaver	OFF	ON / OFF	Moving logo on dark background to reduce LCD wear
Screensaver Timeout	300 seconds	Configurable	Inactivity period before screensaver activates
Full-Screen Alerts	ON	ON / OFF	Full-screen overlay for trips and comm loss events

Home Auto Return: When enabled, the display returns to the Home screen after the configured timeout of inactivity on any settings screen. Recommended for most installations. Disable temporarily during extended service sessions if you need a sub-screen to remain visible.

Logo Screensaver: When enabled, the display shows a slowly moving logo after the configured idle period. Touch anywhere to wake the screen — the first touch only wakes the display and does not activate any button.

Full-Screen Alerts: When enabled, a full-screen overlay appears automatically on trip conditions and communication loss events (see Section 6.4). Disable during active commissioning or on SCADA-supervised sites where the control system handles alarm annunciation. Disabling this setting does not suppress controller trips or Modbus alarm registers.

5.1.2 Security and PIN Lock

The HMI supports an optional PIN lock. When enabled, live data remains visible but configuration changes require entry of a 4-digit PIN. An inactivity timer (default: 30 seconds) can automatically return the display to the Home screen.

If the HMI is unavailable (for example, due to damage or failure), the controller continues to operate using its last saved configuration and can still be monitored/configured via Modbus RTU (RS-485).

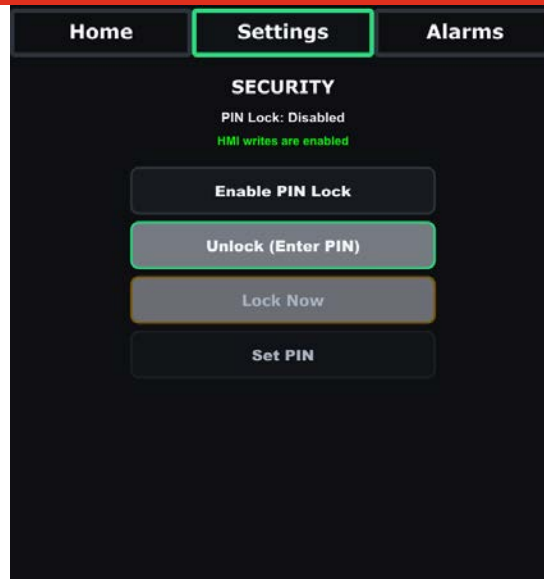


Figure 5.1.2 - Default Security Settings Screen

PIN lock is configured under **Settings** → **Security** on the Dashboard.

When PIN lock is enabled and engaged, the display remains fully functional for viewing data, but all configuration edits are blocked until the correct PIN is entered.

Status	Meaning
PIN Lock: Disabled	No PIN set. All fields editable.
PIN Lock: Enabled (Unlocked)	PIN set; fields editable this session only.
PIN Lock: Enabled (Locked)	PIN active. All fields write-protected.

5.1.3 Setting a PIN for the first time:

1. Navigate to Settings → Security.
2. Confirm status shows **PIN Lock: Disabled**.
3. Tap **Enable PIN Lock**.
4. Enter a 4-digit PIN (0000–9999) on the numeric keypad.
5. Re-enter the same PIN to confirm.
6. Tap **Lock Now** to immediately engage write protection.

Document the PIN in the site maintenance log or commissioning records.

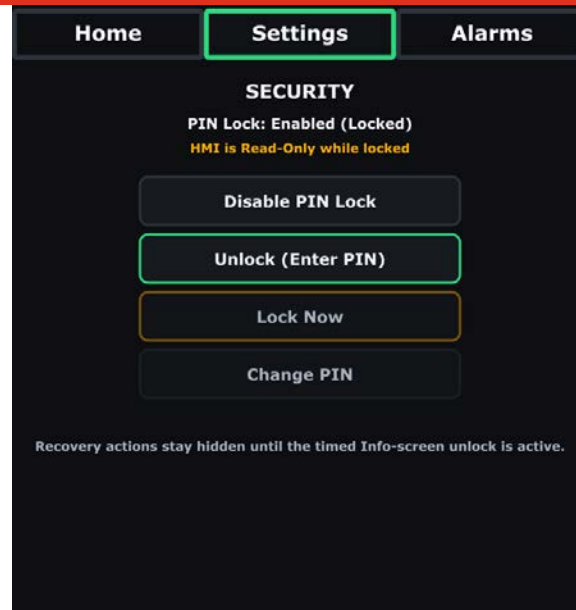


Figure 5.1.4 -Locked Security Settings Screen

5.1.4 Unlocking the interface:

1. Navigate to Settings → Security.
2. Tap **Unlock (Enter PIN)**.
3. Enter the 4-digit PIN and confirm.

The interface remains unlocked for the current power session only. The lock re-engages automatically on the next power cycle.

5.1.5 Changing the PIN:

1. Navigate to Settings → Security → **Change PIN**.
2. Enter the current PIN to verify authorization.
3. Enter and confirm the new 4-digit PIN.

5.1.6 Recovery — forgotten PIN:

1. Navigate to Settings → Info.
2. Tap the firmware version label 10 times within 6 seconds. A countdown indicator appears.
3. Recovery mode activates for 60 seconds and the display navigates to the Security screen.
4. Tap **Reset HMI Security** — clears the PIN and disables PIN lock.

NOTE: *Reset HMI Security only clears the PIN, preserving all other controller settings. It is the recommended recovery method for a forgotten PIN. To erase all setpoints and security settings entirely, perform a **factory reset** (see Section 3.3).*

5.2 Temperature Setpoints

Temperature control parameters determine when the heater output energizes and de-energizes, which sensors contribute to the control decision, and how multiple sensors are combined. Navigate to Control Settings → Temperature Control, or press the “Setpoint” field on the Home Screen. Tap the pencil icon next to the Setpoint field, enter the desired temperature, and confirm. Tap the pencil icon next to the Hysteresis field, enter the deadband value, and confirm. The change takes effect immediately.

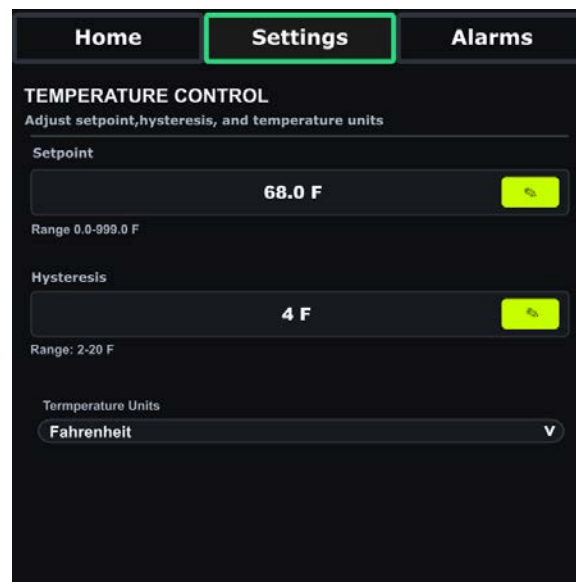


Figure 5.2 - Temperature Control Screen

Parameter	Description	Range	Default
Control Setpoint	Target temperature; output switches at setpoint $\pm$ hysteresis/2	0.0 to 999 °F	68.0 °F
Hysteresis	Deadband around setpoint; prevents rapid cycling	2.0–20.0 °F	4.0 °F
Temperature Units	Display and control units	°F / °C	°F
Multi-RTD Mode	Rule for combining multiple control sensors	1=Low, 2=Average, 3=High	1 (Low)
RTD High Limit	Hard over-temperature cutoff; shuts down heater if exceeded by any control sensor	Configurable (°F)	450.0 °F

The following table describes typical output behavior for setpoints and conditions:

Setpoint	Hysteresis	Output turns ON when	Output turns OFF when
68 °F	4 °F	Temperature falls below 66 °F	Temperature rises above 70 °F
42 °F	6 °F	Temperature falls below 39 °F	Temperature rises above 45 °F

A hysteresis of 2–4 °F is suitable for stable indoor installations. For outdoor freeze protection in rapidly changing conditions, use 6–10 °F to reduce relay cycling and contact wear.

5.2.1 Implementing More than 1 RTD

Navigating to the **RTD Sensors** screen, (Settings > Control Settings > RTD Sensors) users can determine the number of RTD(s) that a controller uses and determine RTD input behavior.

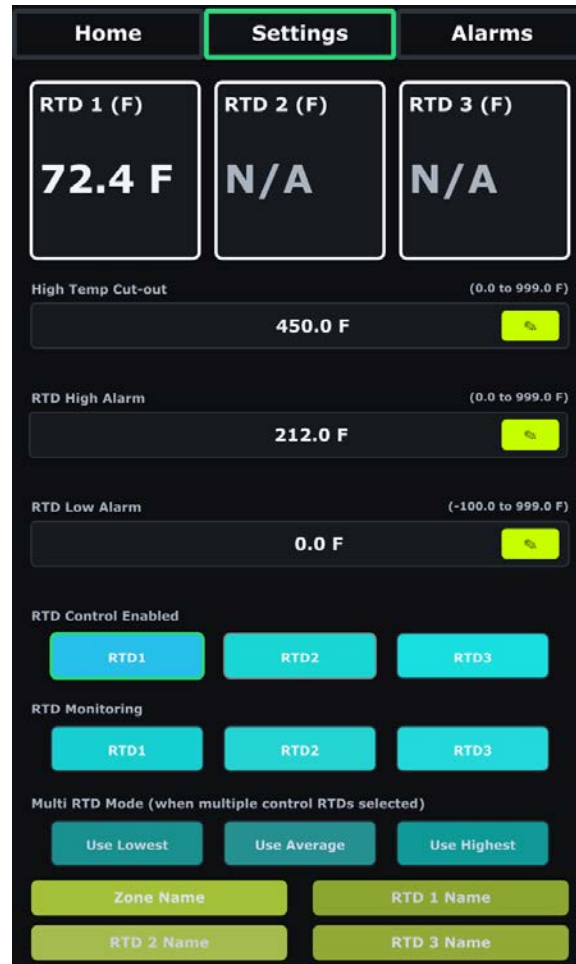


Figure 5.2.2 - RTD Sensor Screen

RTD Control Enabled: Any RTD sensor actively participating in the heating control loop will feature a **bright green border (stroke)** around its respective button (**RTD1**, **RTD2**, or **RTD3**). If there is no green border, that sensor is not influencing the heater output. **RTD Monitoring:** Similarly, a **bright green border** around the buttons in the monitoring row indicates which sensors are actively being tracked for alarms

By default RTD 1 is the primary control sensor and is required. RTD 2 and RTD 3 are optional and can be added to the control mask, monitor mask, both, or neither. Only sensors with a healthy reading contribute — a faulted sensor is automatically excluded from the control calculation.

5.2.2 Multi-RTD Control Modes

When two or more sensors are selected, the Multi-RTD Mode determines how their readings are combined into a single control temperature:

Mode	Control Temperature Used	Typical Application
Lowest	Lowest reading among healthy control sensors	Freeze protection — heater runs until the coldest point is satisfied
Average	Arithmetic mean of healthy control sensors	Balanced dual-sensor control
High	Highest reading among healthy control sensors	Overheat protection — heater turns off if any sensor is hot

NOTE: The Control Temperature is not always the same as any single RTD reading. If multi-RTD mode is configured, it may be the lowest, average, or highest of several sensors. If the displayed Control Temp looks unexpected, navigate to Control Settings → RTD Sensors to observe each RTD sensor individually.

Application guidance:

Application	Recommended Control Mask	Recommended Mode
Single-pipe freeze protection	RTD 1 only (mask 1)	Low
Redundant freeze protection (safety-critical)	RTD 1 + RTD 2 (mask 3)	Low — heater runs until both sensors are satisfied
Balanced dual-sensor control	RTD 1 + RTD 2 (mask 3)	Average
Multi-zone maximum coverage	All three (mask 7)	Low

5.2.3 Renaming RTD Sensors

Each RTD sensor can be assigned a descriptive name displayed on the Dashboard Home screen and in alarm logs. Navigate to **Control Settings** → **RTD Sensors**, tap **RENAME** (see figure 5.2.2) on the desired sensor card, enter the label, and confirm.

5.2.4 Zone Renaming

The control zone can be assigned a descriptive label. The control-point name slot occupies 40125–40141: register 40125 contains the length and 40126–40141 contain packed character data. Use a label that identifies the physical heat-trace circuit.

To rename the zone, navigate to **Control Settings** → **RTD Sensors**, tap **RENAME** (see figure 5.2.2), enter up to 16 characters using the on-screen keyboard, and confirm. The new label takes effect immediately, propagates to the header of the Home screen,

and is included in all subsequent log entries and external interfaces. Zone names persist across power cycles and are preserved through firmware updates; they are erased by a factory reset.

5.3 Temperature Alarms

5.3.1 Types of Temperature Alarms

The controller features two distinct categories of temperature alarms to protect your system:

- **RTD Threshold Alarms** monitor individual sensor health across multi-RTD setups. The RTD Threshold Alarms setpoints are configured on the RTD setting screen (Under **Control Settings** → **RTD Settings**, see figure 5.2.2)
- **Control Temperature Alarms** (which directly safeguard the heating loop).

Each alarm can be independently enabled or disabled, and configured as latching or auto-clearing.

5.3.2 RTD Threshold Temperature Alarms

Parameter	Description	Default	Enabled	Latching
High Temp Cutout	Temperature Setpoint at which if any of the control RTD sensor(s) reaches, will not allow heater to operate.	450.0 °F	Yes	No
RTD High Alarm	Upper RTD Alarm level for individual RTD sensors within multi-RTD monitoring.	212.0 °F	No (Can be enabled in the Settings → Control Settings → Control Alarms)	No
RTD Low Alarm	Lower RTD Alarm level for individual RTD sensors within multi-RTD monitoring.	0.0 °F	No (Can be enabled in the Settings → Control Settings → Control Alarms)	No

Application Guidance:

High Temperature Cutout: Enabled by default as a critical safety layer to prevent a temperature runaway scenario.

RTD High Threshold & RTD Low Threshold Alarms are **DISABLED** by default because they are intended specifically for Multi-RTD zone monitoring. Since the factory default assumes a single RTD input, keep these disabled unless you are monitoring independent zones

Unlike the threshold alarms, the Low and High Control Temperature Alarms are **ENABLED** by default. These function independently and can be customized by navigating to **Settings** → **Control Settings** → **Control Alarms**.

5.3.3 Control Temperature Alarms

Parameter	Description	Default	Enabled	Latching
RTD Failure Alarm	Any control RTD sensor reports an invalid reading	—	Yes	Yes
Low Control Temperature Alarm	Lower Control Temperature Alarm Setpoint for operational alarm.	41.0 °F	Yes	No
High Control Temperature Alarm	Upper Control Setpoint for operational alarm.	212.0 °F	Yes	No
Alarm Relay Settings	Failsafe Mode	OFF	—	—
Alarm Relay Settings	Trip Forces off	OFF	—	—

For freeze-protection circuits, set the Control Temperature Low alarm at 41 °F or slightly higher (45 °F) to provide early warning of a heating failure. Do not disable this alarm.

5.4 Electrical Settings

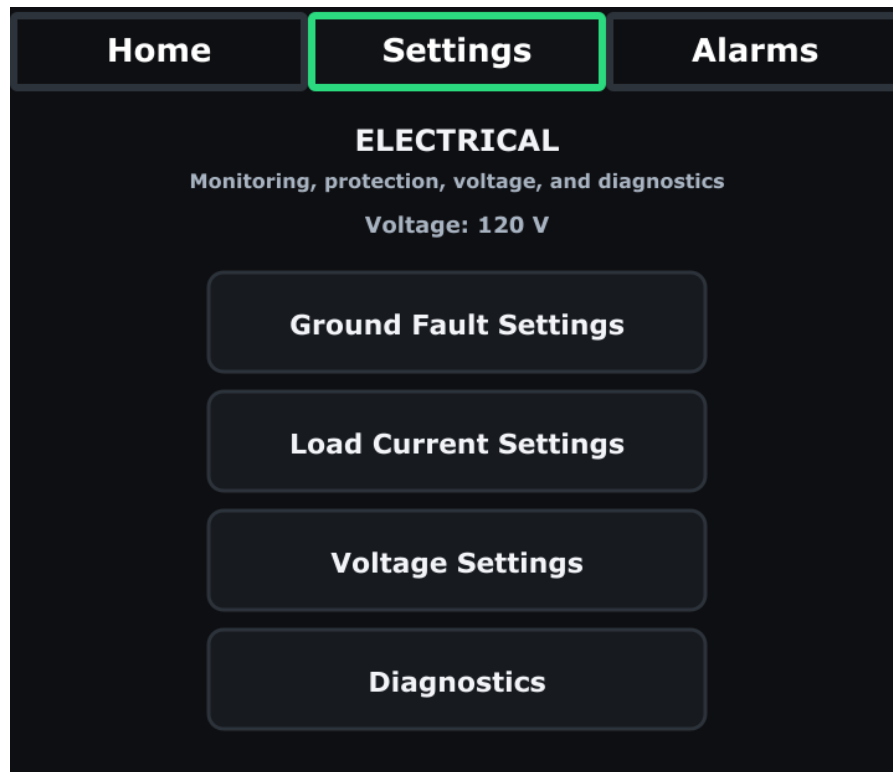


Figure 5.4 Electrical Settings Menu

Electrical Settings define how the controller monitors current, voltage, and relay behavior on the heater circuit. Use this section to set load-current thresholds (low/high alarms and the hard-shutdown trip), ground fault alarm and trip levels, supply voltage alarms, and stuck-relay detection. These protections run continuously during normal operation and are independent of temperature control — a fault here will alarm or trip the heater output regardless of RTD readings. Configure thresholds during commissioning using your measured circuit values; defaults are starting points, not site-specific limits.

5.4.1 Ground Fault Current Protection



Figure 5.4.1 Ground Fault Current Settings

Parameter	Description	Default	Enabled	Latching
GF Alarm enable	Enable/Disable the Ground Fault High Alarm	ON	Yes	N/A
GF alarm latching	Enables Latching of Ground Fault High Alarm	ON	Yes	N/A
Ground Fault — Alarm	Setpoint for Ground Fault High Alarm current	20 mA	Yes	If Enabled
Ground Fault — Trip	Setpoint for Ground Fault Trip current (Must be higher than the Ground Fault Alarm Setpoint)	30 mA	Always	Always
GF Multiplier	Multiplier for GF current calculation (For most applications leave at 1.0)	1.0	N/A	N/A

The ground fault trip threshold must always be set higher than the ground fault alarm threshold. Do not disable ground fault protection without documented site authorization.

5.4.2 Load Current Protection

Home

Settings

Alarms

LOAD CURRENT

Live monitoring and protection settings

Current Load

12.5 A

Load Current Alarms

High alarm enabled	ON	
High alarm latching	OFF	
Low alarm enabled	ON	
Low alarm latching	OFF	
Low alarm threshold	1.0 A	EDIT
High alarm threshold	29.0 A	EDIT
Trip threshold	30.0 A	EDIT
Load Multiplier	1.0 x	EDIT

Stuck Relay Detection

Minimum sensed current	0.5 A	EDIT
Detection delay	320 ms	EDIT

Figure 5.4.2 Load Current Settings

Refer to Figure 5.4.2 – Load Current Settings

Parameter	Description	Default	Enabled	Latching
Load Current High Alarm Enable	Enable / Disable High Load Alarm	ON	Yes	N/A
Load Current High Alarm Latching	When enabled High Load Alarm latches and disables heater operation until reset is cleared	ON	Yes	N/A
Load Current Low Alarm Enable	Enable / Disable Low Load Alarm	ON	Yes	N/A
Load Current Low Alarm Latching	When enabled Low Load Alarm latches and disables heater operation until reset is cleared			
Load Current — Low Alarm Threshold	Lower level value for load current threshold when heater is ON	1.0 A	Yes	No
Load Current — High Alarm High Alarm Threshold	Higher level value for load current threshold when heater is ON	29.0 A	Yes	No
Load Current — Trip	Always enabled, If Current sensed exceeds the trip threshold; hard shutdown, Must be reset.	30.0 A	Always	Always
Load Multiplier	Multiplier for load current calculation (For most applications leave at 1.0)	1.0	N/A	N/A

Via the Dashboard: Settings → Electrical Settings → Ground Fault Settings and Load Current Settings.

Measure actual circuit current during commissioning before finalizing current thresholds. Cold-start current can be significantly higher than steady-state current. Set the low alarm below the minimum expected cold-start draw to avoid nuisance alarms at startup.

5.4.3 Stuck Relay Detection

Refer to Figure 5.4.2 – Load Current Settings

Parameter	Description	Default
Minimum Current	Current level that indicates the relay contacts are closed	0.3 A
Detection Delay	Time window to confirm current is present after relay commanded OFF	500 ms

Stuck relay detection monitors for current flowing when the heater output is commanded off. If current above the minimum threshold persists beyond the detection delay, a Stuck Relay trip is generated. This trips the heater output and requires manual reset.

Do not reduce the detection delay below 200 ms — shorter windows can cause nuisance false trips from normal relay mechanical variance.

5.4.4 Voltage Alarms

The screenshot shows the 'Settings' tab of the HT-ProMon interface. Under the 'VOLTAGE' section, it displays the 'Measured Line Voltage' as 123 V with a status of 'In Range'. The 'Voltage Alarm Configuration' section includes settings for 'Low alarm enabled' (ON), 'Low alarm latching' (OFF), 'Low threshold' (80 V), 'High alarm enabled' (OFF), 'High alarm latching' (OFF), and 'High threshold' (305 V). The 'Voltage Scaling' section shows the 'Voltage Ratio' as 1.0 x. All threshold values have a lock icon next to them.

Figure 5.4.4 Voltage Settings

Voltage control parameters determine if and when end users are alerted to high or low voltage conditions. Navigate to **Control Settings → Electrical Settings → Voltage Settings**, or press the “Voltage” field on the Home Screen.

Parameter	Description	Default	Enabled	Latching
Voltage – Low Alarm Enable	Enable/Disable Low Voltage Alarm	ON	Yes	N/A
Voltage – Low Alarm Latching	Enables Latching of Low Voltage Alarm	ON	Yes	N/A

Voltage — Low Alarm	Supply voltage falls below threshold	85.0 V	Yes	No
Voltage – High Alarm Enable	Enable/Disable High Voltage Alarm	OFF	No	N/A
Voltage – High Alarm Latching	Enables Latching of High Voltage Alarm	OFF	No	N/A
Voltage — High Alarm	Supply voltage exceeds threshold	305.0 V	No	No

The low voltage alarm detects dropped phases, loose connections, or heavily loaded feeders. The high voltage alarm is disabled by default due to susceptibility to normal transient spikes; enable only if the site has a documented overvoltage risk.

Via the Dashboard: Settings → Electrical Settings → Voltage section.

5.6 Output Operating Modes

The HT-ProMon controller operates in one of two output modes. The active mode is visible in the config serial output under OUTPUT CONTROL, and on the Dashboard SSR screens when applicable.

5.6.1 EMR Mode (Electromechanical Relay)

Control Temperature	Heater Output	Alarm Relay
Below (setpoint – hysteresis/2)	ON	OFF (normal)
Above (setpoint + hysteresis/2)	OFF	OFF (normal)
In deadband (between thresholds)	Unchanged	OFF (normal)
RTD fault (invalid reading)	Immediately OFF	ON
RTD high limit exceeded	Immediately OFF	ON
Any trip active	Immediately OFF	ON
Alarm only (no trip)	Normal operation continues	ON

In relay mode, the heater output is binary: fully on or fully off. The control algorithm compares the measured control temperature to the setpoint and switches the relay when the temperature crosses the configured switching thresholds.

1. **Output turns ON** when the control temperature falls below (setpoint – hysteresis/ default = 2).
2. **Output turns OFF** when the control temperature rises above (setpoint + hysteresis/ default = 2).
3. **In deadband** (between the two thresholds): current output state is maintained — no switching occurs.

The Output Control Screen shows only two options when the Output Control screen is displayed: **Minimum on-time** and **Minimum off-time** (default 2000 ms each) prevent rapid cycling when temperature hovers at the switching point. These protect relay contact life regardless of how frequently the temperature crosses the threshold.

5.6.2 SSR Mode (Solid State Relay with Time-Proportional Output)

In SSR mode, the heater output is modulated continuously using a fixed 400 ms cycle period. The PID algorithm calculates a duty cycle between 0% and 100%, and the SSR is switched on for that fraction of each cycle.

4. The current ceiling supervision layer limits the effective duty cycle to keep the load current within the configured ceiling, regardless of what the PID demands.
5. Minimum duty cycles below approximately 4% are automatically rounded down to 0% (fully off) to prevent unreliable operation at very low power levels.
6. Stuck relay detection is not active in SSR mode, because current flows continuously at partial duty when the output is running. In SSR mode, the PID algorithm produces a raw duty cycle demand. The current ceiling supervision layer then evaluates whether executing that duty cycle would exceed the current ceiling:
 7. If estimated current is within the ceiling: the supervised duty equals the raw PID demand.
 8. If estimated current would exceed the ceiling: the supervised duty is capped at the limit that keeps current at the ceiling.

The actual duty cycle applied to the SSR is always the supervised duty. The Dashboard Output Control screen shows both the raw PID demand and the supervised duty in real time, along with a "Limiting" flag when the ceiling is actively reducing the output.

5.7 Output Configuration (SSR Models)

SSR (Solid State Relay) models use a Time-Proportioning Output (TPO) method to control heater power. In this mode, the output is not limited to simple on/off switching. Instead, the controller can apply PID control to calculate a heater duty cycle based on the difference between the current control temperature and the setpoint. That duty cycle is then applied over a fixed 400 ms time window. For example, at 25% output, the SSR is energized for about 100 ms and off for about 300 ms during each 400 ms cycle. This allows the heater power to be regulated smoothly and proportionally, especially as the process temperature approaches the setpoint.

NOTE: The Dashboard Output Options screen routes to Relay Timing parameters in EMR Mode, and to the SSR Output Control screen in SSR Mode. The SSR Output Control screen is hidden when the controller is in EMR Mode.

5.7.1 PID Parameters (SSR Mode)

Home

Settings

Alarms

OUTPUT CONTROL

Advanced SSR output settings and diagnostics.
Visible only when the controller reports SSR mode.

Auto-Tune PID

PID Tuning

Kp	1.00	
Ki	0.00	
Kd	0.00	

Output Limits

Current Ceiling	25.0 A	
-----------------	--------	--

Live Output Telemetry (read-only)

SSR Duty Output	100 %
-----------------	-------

Output Diagnostics (read-only)

0x0004 = CURR-LIM active. At 400 ms this may hold steady while the current ceiling is binding; rapid toggling is not expected.

Raw PID Duty	100.00 %
Supervised Duty	100.00 %
Current Limiting	OK

Figure 5.7.1 Output Control SSR Settings

When using a Solid State Relay, the PID algorithm calculates the heater duty cycle within each 400 ms cycle based on the difference between the current control temperature and the setpoint.

Refer to Figure 5.7.1 – Output Control SSR Settings

Parameter	Description	Default
Kp — Proportional Gain	Response proportional to current temperature error. Higher Kp = more aggressive, faster response, may overshoot.	1.00
Ki — Integral Gain	Correction for persistent steady-state offset. Higher Ki eliminates offset faster; too high causes slow oscillation.	0.00
Kd — Derivative Gain	Predictive brake based on rate of temperature change. Reduces overshoot on fast-rising temperature; too high causes erratic output.	0.00
SSR Current Ceiling	Maximum load current (A); duty is reduced if the estimated RMS current would exceed this value	25.0 A

Via the Dashboard: Settings → Control Settings → Output Options → (SSR mode) Output Control screen.

5.7.3 Current Ceiling

The current ceiling limits the maximum heater duty cycle to keep the estimated load current within a configured limit, regardless of what the PID algorithm demands.

The controller estimates whole-cycle RMS current based on the measured ON-phase current and the current duty. If the estimate would exceed the ceiling, the duty is automatically capped.

Via the Dashboard: Settings Naturalize the use of the Control Settings → Output Options → Output Control screen → Current Ceiling field.

NOTE: Set the current ceiling based on the heater element rating and the circuit breaker rating for the installation. A zero ceiling suppresses all SSR output.

NOTE: If the Dashboard shows "Limiting" in the output diagnostics panel during steady-state operation (not just during cold startup), the current ceiling is actively restraining the output. Either the ceiling is set below the circuit's normal operating current, or the setpoint demand is very high relative to the heater capacity.

5.7.2 SSR Application Output

Gain tuning quick reference:

Symptom	Likely Cause	Adjustment
Temperature never reaches setpoint	Kp or Ki too low	Increase Kp first, then add small Ki
Temperature oscillates above and below setpoint	Kp too high	Reduce Kp by 30–50%
Temperature overshoots setpoint on startup	Kp too high or Kd too low	Reduce Kp or add small Kd
Output is erratic / duty cycle fluctuates rapidly	Kd too high	Reduce or zero Kd
Temperature sluggishly tracks setpoint	All gains too low	Run auto-tune or increase Kp

Starting-point recommendations by application:

Application	Kp	Ki	Kd	Notes
Buried pipe (large thermal mass, slow system)	2.0	0.1	0	Slow system tolerates moderate proportional gain; integral eliminates long-term offset
Exposed instrument tubing (small thermal mass, fast system)	0.5	0.05	0.1	Small Kp to prevent oscillation; small derivative to catch rapid temperature rise

5.7.4 Auto-Tune (SSR Mode)

The auto-tune function automatically calculates Kp and Ki by temporarily driving a controlled oscillation in the heater output and measuring the thermal response. Auto-tune is available in SSR TPO mode only.

Pre-conditions — auto-tune cannot start if any of the following are present:

Blocking Condition	Dashboard Message
Active load current or ground fault trip	"Active trip — clear before tuning"
RTD sensor failure	"RTD not present — RTD required for auto-tune"
Active alarm of any kind	"Active alarm — clear before tuning"

Confirm that no personnel are working on the heated equipment before starting auto-tune — the heater will cycle noticeably during the oscillation phase.

To run auto-tune:

1. Confirm the controller is in SSR TPO mode and all faults and alarms are cleared.
2. Navigate to Settings → Control Settings → Output Options → **Auto-Tune PID**.
3. Verify Status reads "Idle."
4. Tap **Relay Amplitude** and set the desired percentage. Use **20 %** for most installations. Use **10–15 %** for loads above 70 A — at high amplitude, the current ceiling may activate during oscillation and produce incorrect gain results.
5. Tap **Start Auto-Tune**.
6. **SETTLING phase:** The controller waits for stable temperature conditions. This may take several minutes depending on thermal mass.
7. **OSCILLATING phase:** The heater output cycles on and off. Temperature will fluctuate around the setpoint — this is normal.
8. **COMPLETE:** The screen shows calculated Kp and Ki values. Tap **KEEP** to apply, or **REVERT** to restore pre-tune values.

Auto-tune calculates K_p and K_i only. K_d is left at its pre-tune value. Verify K_d manually after accepting auto-tune results if derivative action is in use.

Auto-tune results are a starting point. Monitor the system after applying and make manual adjustments if significant overshoot or slow recovery is observed.

6 Alarm Relay Behavior

The alarm relay output is a separate dry contact from the heater output. It activates on any alarm or trip condition:

Alarm Relay Mode	Behavior
Normal (default)	Relay energizes when any alarm or trip is active; de-energizes when all clear
Failsafe	Relay energizes continuously during normal operation; de-energizes on alarm or trip

The failsafe mode is appropriate when a relay de-energization due to power loss should be treated the same as an alarm condition by the monitoring system.

An additional option causes any trip condition to force the alarm relay to the de-energized state regardless of its configured mode. This is appropriate when the monitoring system is designed to detect a loss of the alarm relay signal as a critical event.

6.1. Alarms vs. Trips

Category	Effect on Heater	Auto-Clears	Requires Manual Reset
Alarm	Heater continues normal operation	Yes (if not set to latching)	Only if latching is enabled
Trip	Heater immediately shuts down and stays off	No	Always required

A trip is a protective shutdown. The controller will not restore heater output until the trip is manually reset. Always identify and correct the root cause of a trip before resetting.

Pressing RESET while the fault condition is still present will momentarily clear the trip flag, but the controller will re-trip immediately when output resumes if the underlying fault persists.

6.1.2 Alarm Types

Alarm	Default Enabled	Default Latching	Condition	Default Limit
Load Current — High	Yes	No	Measured load current exceeds configured threshold	29.0 Amps
Load Current — Low	Yes	No	Measured load current falls below configured threshold when heater is ON	1.0 Amp
Ground Fault — High	Yes	No	Measured leakage current exceeds configured threshold	20 mA
Ground Fault Trip	Yes	Yes	Measured leakage current exceeds the configured threshold	30 mA
RTD Failure	Yes	Yes	Any control RTD sensor reports an invalid reading	
Low Control Temperature Alarm	Yes	No	Control temperature falls below configured threshold	41 F
High Control Temperature Alarm	Yes	No	Control temperature rises above configured threshold	212 F

IMPORTANT — RTD Fault Behavior

An RTD fault de-energizes the heater output and raises the RTD Failure alarm (enabled and latching by default). No trip state or manual reset is required — if the alarm is not latched, the heater resumes when the sensor recovers. See Section 3.2.3 for full behavior and configuration path.

In freeze-protection applications, heater shutdown on RTD fault can result in pipe freeze. Monitor RTD sensor health via the Dashboard or Modbus. Install redundant RTD sensors on critical circuits and configure the control mask to use multiple sensors.

6.1.3 What Happens When an Alarm Activates

1. The alarm relay energizes (or de-energizes in failsafe mode).

2. System Status LED changes to solid amber/yellow.
3. The Dashboard Alarm indicator shows the alarm as ACTIVE (orange).
4. Modbus alarm status registers update immediately.
5. If configured as latching, the alarm remains set even after the condition clears.

6.2 Trip Types

Trip	Condition	Always Latching	Default Limits
Load Current Trip	Measured load current exceeds trip threshold	Yes	30.0 Amps
Ground Fault Trip	Measured leakage current exceeds trip threshold	Yes	30 mA
Stuck Relay	Current detected above minimum when heater is commanded OFF	Yes	300 mA 300ms

6.2.1 What Happens When a Trip Activates

1. The heater output immediately de-energizes and is locked off.
2. The alarm relay energizes (or de-energizes in failsafe mode).
3. LED 0 changes to solid red.
4. The Dashboard displays a Critical Alert overlay (if Full-Screen Alerts are enabled) with a deep red background.
5. Modbus trip status registers update immediately.
6. The trip flag remains set until manually cleared — it does not clear on its own.

6.2.3 Resetting Alarms and Trips

Before resetting, identify and correct the root cause of the fault.

Method	Procedure
Dashboard	Home → Alarms tile → Tap RESET (red button). Or use the Critical Alert overlay → Reset Trip .
Modbus	Write coil 23 (Reset Trip) with value 0xFF00 using function code 05 (See Section 7)

A single reset operation clears all trip flags and all latched alarm flags simultaneously. The Latched Events counter on the Dashboard Alarms screen is cumulative and is cleared by cycling power to the HT-ProMon.

6.2.4 Critical Alert Overlay (Dashboard)

The Dashboard displays a full-screen critical alert automatically when a serious fault occurs and Full-Screen Alerts are enabled in Display Settings.

Condition	Overlay Background
Load Current Trip or Ground Fault Trip	Deep red
Dashboard — Controller communication loss	Deep blue

Button	Action
Acknowledge	Dismisses the overlay and navigates to the Alarms screen for full fault review
Reset Trip	Clears trip flags immediately and returns to the Home screen

Use **Acknowledge** when you need to review the full fault status before clearing. Use **Reset Trip** only after you have confirmed the fault is resolved and it is safe to restore heating. Note that Reset Trip from the overlay clears only trip bits — latched alarms require using the Alarms screen RESET button.

6.2.5 Latched Events Counter

The Dashboard Alarms screen displays a cumulative count of alarm and trip events since the last power-up. This counter is not cleared by the RESET button. A rising count between scheduled service visits — with no current active faults — indicates intermittent events worth investigating:

1. Loose RTD wiring causing momentary sensor failures
2. Cold-start current spikes briefly triggering load alarms
3. Wet-weather ground fault events on aging heater elements

Record the counter at each service visit. A significant increase since the previous visit is diagnostic information warranting attention before the next scheduled inspection.

7. RS-485 / Modbus RTU Reference

The HT-ProMon controller provides two independent Modbus RTU slave ports over RS-485. Both ports share one register space. Either port can read measurements and status or write permitted configuration parameters.

7.1 Port Configuration

Each RS-485 port has its own UART, A/B connections, slave ID, and baud rate. Instance 1 normally serves the local HMI; Instance 2 serves remote monitoring. Connect each master to the appropriate port. The two instances do not share the same A/B terminals.

Default port configuration:

Instance	Default Slave ID	Default Baud Rate
Instance 1 (Primary)	1	115200
Instance 2 (Secondary)	1	9600

Frame parameters (fixed):

Parameter	Value
Data bits	8
Parity	None
Stop bits	1
Flow control	None

7.2 Protocol Summary

The controller responds to standard Modbus RTU function codes:

Function Code	Name	Used For
FC01	Read Coils	Read coils 00001–00038
FC03	Read Holding Registers	Read holding registers 40001–40192
FC04	Read Input Registers	Read input registers 30001–30038
FC05	Write Single Coil	Trigger write-edge coils (00023 reset, 00024 factory reset)
FC06	Write Single Holding Register	Write a single configuration register
FC15	Write Multiple Coils	Write multiple coils
FC16	Write Multiple Holding Registers	Write multiple configuration registers

Temperature input registers 30001, 30003, 30005, and 30007 use signed 16-bit values scaled $\times 10$ in °F. Other registers use their separately documented units and scaling.

7.3 Register Map

The register map is organized into three address spaces: input registers (read-only measurements and status, FC04), holding registers (configuration, FC03/06/16), and coils (binary state and commands, FC01/05/15). All multi-byte values use Modbus standard big-endian byte order within each 16-bit register.

7.3.1 Input Registers (30001–30038)

Read with FC04. All input registers are read-only telemetry from the controller. Scaled integer

values use the indicated multiplier (e.g., "×10" means the value is stored as ten times the real value).

Temperatures and electrical measurements:

Address	Register	Scale	Description
30001	REG_RTD1_TEMP_F	×10 °F	RTD 1 temperature
30002	REG_RTD1_RESISTANCE	×10 Ω	RTD 1 resistance
30003	REG_RTD2_TEMP_F	×10 °F	RTD 2 temperature
30004	REG_RTD2_RESISTANCE	×10 Ω	RTD 2 resistance
30005	REG_RTD3_TEMP_F	×10 °F	RTD 3 temperature
30006	REG_RTD3_RESISTANCE	×10 Ω	RTD 3 resistance
30007	REG_CONTROL_TEMP	×10 °F	Calculated control temperature
30008	REG_LOAD_CURRENT	×100 A	Heater load current
30009	REG_GF_CURRENT	×10 mA	Ground-fault current
30010	REG_LINE_VOLTAGE	×100 V	RMS line voltage
30011	REG_LINE_POWER	W	Calculated power

Core status registers:

Address	Register	Description
30012	REG_SYSTEM_STATUS	bit0 heater on, bit1 Wi-Fi connected, bit2 BLE connected
30013	REG_ALARM_STATUS	bit0 high load, bit1 low load, bit2 high GF, bit3 CTRL_TEMP low, bit4 CTRL_TEMP high
30014	REG_TRIP_STATUS	bit0 high load trip, bit1 high GF trip
30015	REG_ALARM_ENABLE_STATUS	bit0 low load EN, bit1 high load EN, bit2 high GF EN, bit3 RTD failure EN, bit4 CTRL_TEMP low EN, bit5 CTRL_TEMP high EN, bit6 RTD low EN, bit7 RTD high EN
30016	REG_ALARM_ACTIVE_STATUS	Live active alarm bitfield. Bits 0–7 match 30015; bit8 voltage low, bit9 voltage high, bit12 resistance low, bit13 resistance high; bits 10–11 and 14–15 reserved

BLE identifier (16 ASCII characters, packed two per register):

Address Range	Register	Description
30017–30024	REG_BLE_ID_1 ... REG_BLE_ID_8	BLE identifier, 16 ASCII chars, two per register

Diagnostics and counters:

Address	Register	Description
30025	REG_VOLT_ALARM_STATUS	Voltage alarm status bitfield
30026	REG_RES_EST_OHM_X10	Estimated heater resistance, ×10 Ω
30027	REG_RES_STATUS	Resistance status bitfield
30028	REG_SELFTEST_STATUS	Self-test status bitfield
30029	REG_SELFTEST_LAST_REASON	Last self-test reason/result

30030	REG_SELFTEST_LAST_RUN_HOURS_LOW	Last self-test uptime hours, low word
30031	REG_SELFTEST_LAST_RUN_HOURS_HIGH	Last self-test uptime hours, high word
30032	REG_RELAY_ON_COUNT_LOW	Relay ON count, low word
30033	REG_RELAY_ON_COUNT_HIGH	Relay ON count, high word
30034	REG_RELAY_OFF_COUNT_LOW	Relay OFF count, low word
30035	REG_RELAY_OFF_COUNT_HIGH	Relay OFF count, high word
30036	REG_RELAY_CYCLE_COUNT_LOW	Relay cycle count, low word
30037	REG_RELAY_CYCLE_COUNT_HIGH	Relay cycle count, high word
30038	REG_RELAY_COUNTER_DIRTY	Relay counter unsaved flag

7.3.2 Holding Registers (40001–40192)

Read holding registers with FC03 and write permitted registers with FC06 or FC16. Addresses 40001–40067 form the frozen inherited base block; 40068–40079 cover output control; 40080–40082 cover auto-tune control; 40083–40094 report LoRa settings; 40095 reports device options; 40096 reports RTD limit inhibition; 40097–40109 report auto-tune status and history; 40110–40119 are reserved; 40120–40122 report firmware version; 40123–40124 are reserved; 40125–40192 hold name strings.

Invalid holding-register writes are rejected before any register updates. FC06 rejections return exception function code 0x86; FC16 rejections return 0x90; both carry exception code 0x03 (Illegal Data Value).

Group 1 — Control Settings (40001–40008)

Address	Register	Scale	Description
40001	HREG_SETPOINT	×10 °F	Temperature setpoint
40002	HREG_HYSTERESIS	×10 °F	Hysteresis
40003	HREG_RTD_MODE	—	Control RTD mask, valid write range 1–7
40004	HREG_MON_MODE	—	Monitor RTD mask, valid write range 0–7
40005	HREG_DUAL_RTD_MODE	—	1=Low, 2=Avg, 3=High
40006	HREG_RTD_LIMIT	×10 °F	High-temperature cut-out threshold
40007	HREG_RTD_ALARM_HIGH	×10 °F	RTD threshold high alarm (control + monitor RTDs)
40008	HREG_RTD_ALARM_LOW	×10 °F	RTD threshold low alarm (control + monitor RTDs)

RTD mask values for HREG_RTD_MODE and HREG_MON_MODE (bitmask: bit0=RTD1, bit1=RTD2, bit2=RTD3):

0 — none (valid only for monitor mode)

1 — RTD1 only

2 — RTD2 only

3 — RTD1 + RTD2

4 — RTD3 only

5 — RTD1 + RTD3

6 — RTD2 + RTD3

7 — RTD1 + RTD2 + RTD3

Write validation enforces $RTD_ALARM_LOW \leq RTD_ALARM_HIGH \leq RTD_LIMIT$.

Group 2 — Protection Thresholds (40009–40015)

Address	Register	Scale	Description
40009	HREG_LOAD_ALARM_LOW	×100 A	Low load alarm threshold
40010	HREG_LOAD_ALARM_HIGH	×100 A	High load alarm threshold
40011	HREG_LOAD_TRIP	×100 A	Load trip threshold (accepted: 10–8000, i.e. 0.10–80.00 A)
40012	HREG_GF_ALARM_HIGH	mA	Ground-fault alarm threshold
40013	HREG_GF_TRIP	mA	Ground-fault trip threshold
40014	HREG_STUCK_RELAY_MIN	×100 A	Stuck-relay current threshold
40015	HREG_STUCK_RELAY_MS	ms	Stuck-relay delay

Group 3 — Alarm Management (40016–40033)

Each alarm condition has a separate enable bit and a separate latch bit. Disabling an alarm (writing 0 to its enable register) clears any pending latch for that alarm. Latched alarms remain asserted after the condition clears, until either reset via COIL_RESET_TRIP or power cycle.

Alarm enable registers (write 1 to enable, 0 to disable):

Address	Register	Description
40016	HREG_LOAD_LOW_ALARM_EN	Enable low load alarm
40017	HREG_LOAD_HIGH_ALARM_EN	Enable high load alarm
40018	HREG_GF_HIGH_ALARM_EN	Enable high ground-fault alarm
40019	HREG_RTD_FAILURE_ALARM_EN	Enable RTD failure alarm
40020	HREG_RTD_LOW_ALARM_EN	Enable RTD threshold low alarm
40021	HREG_RTD_HIGH_ALARM_EN	Enable RTD threshold high alarm
40022	HREG_CONTROL_TEMP_LOW_ALARM_EN	Enable control-temperature low alarm
40023	HREG_CONTROL_TEMP_HIGH_ALARM_EN	Enable control-temperature high alarm

Latching configuration:

Address	Register	Description
40024	HREG_LOAD_LOW_LATCH	Latch low-load alarm (0 = auto-clear, 1 = latch)
40025	HREG_LOAD_HIGH_LATCH	Latch high-load alarm
40026	HREG_GF_HIGH_LATCH	Latch high-GF alarm
40027	HREG_RTD_FAILURE_LATCH	Latch RTD failure alarm
40028	HREG_RTD_LOW_LATCH	Latch RTD threshold low alarm
40029	HREG_RTD_HIGH_LATCH	Latch RTD threshold high alarm
40030	HREG_CONTROL_TEMP_LOW_LATCH	Latch control-temperature low alarm
40031	HREG_CONTROL_TEMP_HIGH_LATCH	Latch control-temperature high

		alarm
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Control-temperature alarm thresholds:

Address	Register	Scale	Description
40032	HREG_CONTROL_TEMP_ALARM_LOW	×10 °F	Control-temperature low alarm threshold
40033	HREG_CONTROL_TEMP_ALARM_HIGH	×10 °F	Control-temperature high alarm threshold

Group 4 — Advanced Sensing (40034–40040)

Address	Register	Scale	Description
40034	HREG_LOW_LOAD_MULT	×100	Low-current mode multiplier
40035	HREG_LOW_LOAD_MIN	×100 A	Low-current entry threshold
40036	HREG_LOW_LOAD_MAX	×100 A	Low-current exit threshold
40037	HREG_PEAK_MARGIN	×100	Peak margin over RMS
40038	HREG_VOLT_RATIO	×100	Voltage ratio
40039	HREG_LOAD_CURR_MULT	×100	Load-current multiplier
40040	HREG_GF_CURR_MULT	×100	Ground-fault multiplier

Behavior note: RTD_ALARM_LOW / HIGH evaluate each configured RTD independently and do not use the calculated control temperature. CONTROL_TEMP_ALARM_LOW / HIGH evaluate the calculated control temperature only and do not evaluate individual RTDs.

Group 5: System, Diagnostics, and Self-Test

RS-485 instances and relay timing (40041–40048):

Address	Register	Description
40041	HREG_SLAVE_ID	Modbus slave ID, instance 1 (1–247)
40042	HREG_BAUDRATE	Baud rate / 100, instance 1
40043	HREG_SLAVE_ID_2	Modbus slave ID, instance 2 (1–247)
40044	HREG_BAUDRATE_2	Baud rate / 100, instance 2
40045	HREG_ALARM_RELAY_FAILSAFE	Alarm relay failsafe mode
40046	HREG_ALARM_RELAY_TRIP_OFF	Trip forces alarm relay off
40047	HREG_RELAY_MIN_ON_MS	Minimum heater ON time, ms
40048	HREG_RELAY_MIN_OFF_MS	Minimum heater OFF time, ms

Voltage alarm thresholds and configuration:

Address	Register	Scale	Description
40049	HREG_VOLT_ALARM_LOW	×100 V	Voltage low alarm threshold
40050	HREG_VOLT_ALARM_HIGH	×100 V	Voltage high alarm threshold
40051	HREG_VOLT_LOW_ALARM_EN	—	Enable voltage low alarm

40052	HREG_VOLT_HIGH_ALARM_EN	—	Enable voltage high alarm
40053	HREG_VOLT_LOW_LATCH	—	Voltage low alarm latching
40054	HREG_VOLT_HIGH_LATCH	—	Voltage high alarm latching

Validation: HREG_VOLT_ALARM_LOW and HREG_VOLT_ALARM_HIGH accept up to 30500 raw (305.00 V). If the HAS_CTRL_XFMR device option is set, the accepted ceiling rises to 60000 raw (600.00 V).

Resistance estimate and alarm configuration:

Address	Register	Scale	Description
40055	HREG_RES_NOMINAL	×10 Ω	Nominal heater resistance
40056	HREG_RES_LOW	×10 Ω	Low resistance threshold
40057	HREG_RES_HIGH	×10 Ω	High resistance threshold
40058	HREG_RES_MIN_CURRENT	×100 A	Minimum current required for resistance estimate
40059	HREG_RES_ALARM_EN	—	Enable resistance alarm
40060	HREG_RES_ALARM_LATCH	—	Resistance alarm latching

Self-test, persistence, and protection sensitivity:

Address	Register	Description
40061	HREG_SELFTEST_EN	Enable scheduled self-test
40062	HREG_SELFTEST_PERIOD_H	Self-test period (hours)
40063	HREG_SELFTEST_ON_MIN	Self-test ON duration (minutes)
40064	HREG_SELFTEST_NOW_CMD	Run self-test immediately (write 1 to trigger; write-edge)
40065	HREG_RELAY_COUNT_SAVE_DELTA	Relay-counter save threshold (cycles between flash writes)
40066	HREG_RELAY_COUNT_RESET_CMD	Reset relay counters (write 1 to trigger; write-edge)
40067	HREG_PROT_SENS	Protection sensitivity factor (×100)

Appended Output-Control Block: 40068–40079

Registers specific to the HT-ProMon SSR output-control path. Mode selection (40068) and SSR cycle period (40069) are factory set and read-only via Modbus.

Address	Register	Access	Scale	Description
40068	HREG_OUTPUT_MODE	RO	1	Active output mode: 0=relay, 1=SSR TPO
40069	HREG_SSR_CYCLE_PERIOD_MS	RO	ms ÷ 100	Fixed 400 ms cycle, reads as 4
40070	HREG_PID_KP	RW	×100	PID proportional gain
40071	HREG_PID_KI	RW	×100	PID integral gain
40072	HREG_PID_KD	RW	×100	PID derivative gain
40073	HREG_SSR_DUTY_OUTPUT	RO	%	Live SSR duty output

40074	HREG_SSR_STARTUP_MAX_DUTY	Reserved	—	Always 0; writes rejected
40075	HREG_SSR_STARTUP_WARMUP_S	Reserved	—	Always 0; writes rejected
40076	HREG_SSR_CURRENT_CEIL_A	RW	×100 A	Current ceiling for SSR supervision
40077	HREG_SSR_PID_RAW_DUTY_X100	RO	×100 %	Raw PID demand
40078	HREG_SSR_SUPERVISED_DUTY_X100	RO	×100 %	Supervised duty command
40079	HREG_SSR_SUPERVISION_FLAGS	RO	bitfield	Supervisor flags (0x0004 = current limiting active)

Behavior: 40068 reports the active output mode and rejects Modbus writes. 40069 is fixed at 400 ms and rejects writes. 40073 and 40077–40079 are live read-only telemetry. 40074–40075 are reserved continuity addresses and reject writes. Supervisor flag bit 0x0004 indicates current limiting is active. 40070–40072 are writable PID terms.

PID Auto-Tune Block (40080–40082)

Auto-tune is only valid in SSR TPO mode. All three registers reject Modbus writes if the controller is currently in relay mode.

Address	Register	Access	Scale	Description
40080	HREG_AUTOTUNE_CONTROL	R/W	1	Write 1 to start auto-tune, 0 to abort
40081	HREG_AUTOTUNE_STATUS	RO	1	0=idle, 1=running, 2=complete, 3=failed
40082	HREG_AUTOTUNE_RELAY_AMP_X100	R/W	%	Relay amplitude, accepts 10–100 (direct percentage)

Read-Only LoRa Block (40083–40094)

Read-only on Modbus; LoRa changes are made through USB serial or BLE. These registers expose the controller's local LoRa configuration and key identity, not receiver-side state.

Address	Register	Scale	Description
40083	HREG_LORA_STATUS_FLAGS	bitfield	0x0001 enabled, 0x0002 initialized, 0x0004 key present, 0x0008 factory default key
40084	HREG_LORA_DEVICE_ID	1	LoRa device ID
40085	HREG_LORA_FREQ_X10	MHz ×10	Frequency
40086	HREG_LORA_BAND_X10	kHz ×10	Bandwidth
40087	HREG_LORA_SPREAD	1	Spreading factor
40088	HREG_LORA_TX_POWER_DBM	signed dBm	TX power

40089	HREG_LORA_UPLINK_INTERVAL_S_LOW	seconds	Uplink interval, low word
40090	HREG_LORA_UPLINK_INTERVAL_S_HIGH	seconds	Uplink interval, high word
40091	HREG_LORA_NET_ID	1	Network ID
40092	HREG_LORA_SEND_DELAY_MS	ms	Legacy send delay
40093	HREG_LORA_KEY_FINGERPRINT_LOW	1	Low word of 32-bit local key fingerprint
40094	HREG_LORA_KEY_FINGERPRINT_HIGH	1	High word of 32-bit local key fingerprint

Device Options, Cutout Status, and Version (40095–40122)

These registers expose installer-set hardware capability flags, the active RTD high-temperature cutout status, and firmware version. Address 40095 (HREG_DEVICE_OPTIONS) is read-only over Modbus; both option bits are set only via USB serial commands (has_contactor on|off, has_ctrl_xfmr on|off) and survive a factory reset.

Device options register:

Address	Register	Access	Description
40095	HREG_DEVICE_OPTIONS	RO (Modbus)	Installer-set hardware capability bitfield; Modbus writes return exception 3

RTD limit cutout status:

Address	Register	Access	Description
40096	HREG_RTD_LIMIT_ACTIVE	RO	1 when RTD high-temperature cutout is actively inhibiting output; firmware applies a short debounce before state changes

PID auto-tune extended status and history occupy 40097–40109 and are read-only. Register 40097 reports the detailed phase (0=idle, 1=settling, 2=oscillating, 3=complete, 4=failed); this encoding differs from the legacy status at 40081. Register 40100 is the live failure reason retained only since boot. Register 40109 is the persisted failure reason of the last completed run. Addresses 40110–40119 and 40123–40124 remain reserved.

Address	Register	Access	Scale	Description
40097	HREG_AUTOTUNE_PHASE	RO	1	0 idle; 1 settling; 2 oscillating; 3 complete; 4 failed
40098	HREG_AUTOTUNE_CROSSING_COUNT	RO	1	Crossings recorded this run
40099	HREG_AUTOTUNE_MIN_CROSSINGS	RO	1	Crossings required for completion
40100	HREG_AUTOTUNE_FAIL_REASON	RO	1	Live stop reason; RAM only, resets on reboot
40101	HREG_AUTOTUNE_OSC_TIMEOUT_S	RO	seconds	Oscillation-phase timeout
40102	HREG_AUTOTUNE_LAST_RUN_OUTCOME	RO	1	Persisted: 0 never; 1 complete; 2

Address	Register	Access	Scale	Description
				failed; 3 aborted
40103	HREG_AUTOTUNE_LAST_RUN_SETPPOINT	RO	×10 °F	Persisted starting setpoint; signed 16-bit
40104	HREG_AUTOTUNE_LAST_RUN_RELAY_AMP	RO	%	Persisted relay amplitude; direct percentage
40105	HREG_AUTOTUNE_LAST_RUN_KP	RO	×100	Persisted resulting Kp
40106	HREG_AUTOTUNE_LAST_RUN_KI	RO	×100	Persisted resulting Ki
40107	HREG_AUTOTUNE_LAST_RUN_HOURS_LOW	RO	hours	Persisted completion uptime, low word
40108	HREG_AUTOTUNE_LAST_RUN_HOURS_HIGH	RO	hours	Persisted completion uptime, high word
40109	HREG_AUTOTUNE_LAST_RUN_FAIL_REASON	RO	1	Persisted last-completed-run stop reason; survives reboot

Failure reason codes for 40100 and 40109: 0 none; 1 start rejected in relay mode; 2 start rejected with active trip; 3 start rejected with RTD failure; 4 start rejected with blocking alarm; 5 aborted on invalid control temperature; 6 aborted on trip; 7 aborted on RTD limit; 8 aborted by user; 9 crossing timeout; 10 result computation failed. Codes 11 (settle timeout) and 12 (setpoint changed) are reserved in this source revision.

Firmware version:

Address	Register	Access	Description
40120	HREG_FW_VERSION_MAJOR	RO	Firmware major version (currently 2)
40121	HREG_FW_VERSION_MINOR	RO	Firmware minor version (reviewed source: 2)
40122	HREG_FW_VERSION_PATCH	RO	Firmware patch version (reviewed source: 3)

Firmware version registers are read-only compile-time values; this reviewed firmware reports 2.2.3. Read deployed version registers rather than assuming this version. An FC03 request at offset 67 (40068), quantity 57, covers 40068–40124. A request at offset 0, quantity 124, covers 40001–40124. Read the name slots separately.

Name String Registers (40125–40192)

Each label supports up to 32 printable ASCII characters (codes 32–126). Each 17-register slot contains a length register followed by 16 packed character registers. The first character is the high byte and the second is the low byte. Unused bytes are zero-filled. Firmware removes unsupported characters and trailing spaces.

Address Range	Register Group	Description
40125–40141	HREG_CTRL_NAME	Control-point label (length + 16 packed character registers)
40142–40158	HREG_RTD1_NAME	RTD 1 label (length + 16 packed character registers)
40159–40175	HREG_RTD2_NAME	RTD 2 label (length + 16 packed

		character registers)
40176–40192	HREG_RTD3_NAME	RTD 3 label (length + 16 packed character registers)

7.3.3 Coils (00001–00038)

Read coils with FC01. Coils marked Write-edge accept a one-shot trigger via FC05 (Write Single Coil) with value 0xFF00 — these are momentary actions, not persistent states. All other coils are read-only status flags maintained by the firmware.

Address	Coil	Access	Description
00001	COIL_HEATER_ON	RO	Heater output ON
00002	COIL_IN_ALARM	RO	Aggregate alarm active
00003	COIL_IN_TRIP	RO	Aggregate trip active
00004	COIL_STUCK_RELAY	RO	Stuck relay active
00005	COIL_RTD1_OK	RO	RTD 1 healthy (1 = no fault)
00006	COIL_RTD2_OK	RO	RTD 2 healthy
00007	COIL_RTD3_OK	RO	RTD 3 healthy
00008	COIL_LOW_LOAD_ALARM	RO	Low load alarm active
00009	COIL_HIGH_LOAD_ALARM	RO	High load alarm active
00010	COIL_HIGH_GF_ALARM	RO	High GF alarm active
00011	COIL_RTD_FAILURE_ALARM	RO	RTD failure alarm active
00012	COIL_CONTROL_TEMP_LOW_ALARM	RO	Control-temperature low alarm active
00013	COIL_CONTROL_TEMP_HIGH_ALARM	RO	Control-temperature high alarm active
00014	COIL_HIGH_LOAD_TRIP	RO	Load trip active
00015	COIL_HIGH_GF_TRIP	RO	GF trip active
00016	COIL_LOAD_LOW_LATCHED	RO	Low load alarm latched
00017	COIL_LOAD_HIGH_LATCHED	RO	High load alarm latched
00018	COIL_GF_HIGH_LATCHED	RO	High GF alarm latched
00019	COIL_RTD_FAILURE_LATCHED	RO	RTD failure alarm latched
00020	COIL_CONTROL_TEMP_LOW_LATCHED	RO	Control-temperature low alarm latched
00021	COIL_CONTROL_TEMP_HIGH_LATCHED	RO	Control-temperature high alarm latched
00022	COIL_STUCK_RELAY_LATCHED	RO	Stuck relay latched
00023	COIL_RESET_TRIP	Write-edge	Reset all alarm/trip latches. Write 0xFF00 to trigger.
00024	COIL_FACTORY_RESET	Write-edge	Trigger factory reset. Write 0xFF00 to trigger.

00025	COIL_VOLT_LOW_ALARM	RO	Voltage low alarm active
00026	COIL_VOLT_HIGH_ALARM	RO	Voltage high alarm active
00027	COIL_VOLT_LOW_LATCHED	RO	Voltage low alarm latched
00028	COIL_VOLT_HIGH_LATCHED	RO	Voltage high alarm latched
00029	COIL_RES_LOW_ALARM	RO	Resistance low alarm active
00030	COIL_RES_HIGH_ALARM	RO	Resistance high alarm active
00031	COIL_RES_LATCHED	RO	Resistance alarm latched
00032	COIL_SELFTEST_ACTIVE	RO	Self-test running
00033	COIL_SELFTEST_PASS_LAST	RO	Last self-test passed
00034	COIL_SELFTEST_FAIL_LAST	RO	Last self-test failed
00035	COIL_RTD_LOW_ALARM	RO	RTD threshold low alarm active
00036	COIL_RTD_HIGH_ALARM	RO	RTD threshold high alarm active
00037	COIL_RTD_LOW_LATCHED	RO	RTD threshold low alarm latched
00038	COIL_RTD_HIGH_LATCHED	RO	RTD threshold high alarm latched

Alarm behavior: *COIL_CONTROL_TEMP_** coils follow the calculated control temperature only. *COIL_RTD_** coils follow per-RTD threshold evaluation across the configured control + monitor RTD set.

7.4 Reading Data

Wire addresses are zero-based within each address space: subtract 30001 for input registers, 40001 for holding registers, and 1 for coils. For example, holding register 40070 uses offset 69 (0x0045). Bytes within each register are transmitted most significant byte first. For low-word/high-word pairs, reconstruct the 32-bit value as low | (high << 16).

Example 1: Read all three RTD temperatures (FC04)

Read input registers 30001, 30003, 30005 in one transaction. Start address on the wire is 0x0000, quantity is 6 (covers RTD1 temp, RTD1 res, RTD2 temp, RTD2 res, RTD3 temp, RTD3 res).

Field	Value
Slave ID	0x01
Function Code	0x04 (Read Input Registers)
Starting Address	0x0000 (= 30001 zero-based)
Quantity	0x0006 (six registers)
CRC	computed by master

Sample response payload: 0x05 0x36 0x04 0x90 0x05 0x16 0x04 0x90 0x04 0xD2 0x04 0x90
→ RTD1 = 1334 (133.4°F), RTD2 = 1302 (130.2°F), RTD3 = 1234 (123.4°F). Resistance words in this example are placeholders (0x0490 = 116.8 Ω).

Example 2: Read PID gains (FC03)

Read PID terms at 40070–40072 (Kp, Ki, Kd) in one FC03 transaction. Each gain is stored $\times 100$.

Request frame:

Field	Value
Slave ID	0x01
Function Code	0x03 (Read Holding Registers)
Starting Address	0x0045 (= 40070 zero-based, decimal 69)
Quantity	0x0003 (three registers)
CRC	computed by master

Sample response payload: 0x00 0x96 0x00 0x32 0x00 0x14 → Kp = 150 (1.50), Ki = 50 (0.50), Kd = 20 (0.20).

Example 3: Read alarm and trip status coils (FC01)

Read the first 8 coils (00001–00008) in one transaction to check heater state, aggregate alarm and trip state, stuck-relay diagnostic, and the three RTD health flags.

Request frame:

Field	Value
Slave ID	0x01
Function Code	0x01 (Read Coils)
Starting Address	0x0000 (= 00001 zero-based)
Quantity	0x0008 (eight coils)
CRC	computed by master

Sample response data byte: 0x71 → bits 0, 4, 5, 6 set. Decoded LSB-first: heater ON (00001), RTD1 OK (00005), RTD2 OK (00006), RTD3 OK (00007). The aggregate alarm and trip coils are clear and there is no stuck-relay diagnostic. Modbus packs coil responses LSB first, so bit 0 of the byte is the lowest-addressed coil in the request.

7.5 Writing Configuration

Configuration changes use FC06 (single holding register), FC16 (multiple holding registers), or FC05 (single coil for write-edge commands). The controller validates every holding-register write before applying it and persists accepted changes through the firmware save flow.

Example 1: Set the temperature setpoint (FC06)

Set the temperature setpoint at 40001 to 75.0 °F. HREG_SETPOINT is scaled $\times 10$, so 75.0 °F is written as raw value 750 (0x02EE).

Request frame:

Field	Value
Slave ID	0x01
Function code	0x06 (Write Single Holding Register)
Register address	0x0000 (= 40001 zero-based)
Register value	0x02EE (decimal 750 → 75.0 °F)
CRC-16	auto

On success the slave echoes the request back. If the value is out of range or violates a cross-register constraint, the controller returns exception function code 0x86 with exception code 0x03 (Illegal Data Value) and no value is written.

Example 2: Write multiple PID gains in one transaction (FC16)

Use FC16 (Write Multiple Holding Registers) to set $K_p = 1.50$, $K_i = 0.50$, and $K_d = 0.20$ at 40070–40072 atomically. Each gain is scaled $\times 100$ ($1.50 \rightarrow 150$, $0.50 \rightarrow 50$, $0.20 \rightarrow 20$).

Request frame:

Field	Value
Slave ID	0x01
Function code	0x10 (FC16, Write Multiple Holding Registers)
Starting address	0x0045 (decimal 69 $\rightarrow$ register 40070)
Quantity of registers	0x0003 (3 registers)
Byte count	0x06 (6 data bytes)
Data payload	0x00 0x96 0x00 0x32 0x00 0x14 ($K_p=150$, $K_i=50$, $K_d=20$)
CRC-16	auto

FC16 is atomic across the entire write — if any one register fails validation, the controller rejects the transaction and returns exception function code 0x90 and exception code 0x03 (Illegal Data Value); no register is updated. Use FC16 whenever related parameters must move together.

Example 3: Clear latched alarms (FC05)

Reset all latched alarm and trip flags by writing the COIL_RESET_TRIP coil at 00023. This is a write-edge coil — write 0xFF00 (Modbus standard "coil ON") to trigger the reset; the controller performs the one-shot action and the coil immediately returns to 0.

Request frame:

Field	Value
Slave ID	0x01
Function code	0x05 (Write Single Coil)
Coil address	0x0016 (= 00023 zero-based)
Coil value	0xFF00 (trigger)
CRC-16	auto

The slave echoes the request on success. The reset clears every latched alarm and trip flag in one operation; coils that represent live (non-latched) conditions are not affected and will re-assert immediately if the underlying fault is still present.

Write Validation and Cautions

Holding-register writes are validated before any register is updated. FC06 rejections return exception function code 0x86; FC16 rejections return 0x90. Both use exception code 0x03 (Illegal Data Value). When FC16 rejects a write, none of the registers in the request are updated — the operation is atomic.

Cross-register constraints are enforced at write time. For example, the firmware rejects any write that would result in RTD_ALARM_LOW being greater than RTD_ALARM_HIGH or RTD_ALARM_HIGH being greater than RTD_LIMIT, even if each individual value is in the controller's allowable input range. When writing related thresholds, send them in a single FC16 transaction so the new values are evaluated together.

Write-edge coils (COIL_RESET_TRIP at 00023, COIL_FACTORY_RESET at 00024) and write-edge holding commands (HREG_SELFTEST_NOW_CMD at 40064, HREG_RELAY_COUNT_RESET_CMD at 40066) are not persistent settings — writing 1 or 0xFF00 triggers a one-shot action and the value immediately returns to 0. Reading these

registers always returns 0.

Caution: COIL_FACTORY_RESET (00024) erases all controller configuration — setpoints, thresholds, alarm enables, calibration multipliers, slave ID and baud rate, and name strings — and reverts the controller to firmware defaults.

The DEVICE_OPTIONS flags (HAS_CONTACTOR, HAS_CTRL_XFMR) survive a factory reset because they describe installed hardware. This coil cannot be written via routine integration.

8. Troubleshooting

8.1 Symptom and Corrective Action Reference

The following table covers both temperature/sensor faults and communication faults. For field verification of sensor resistance values, refer to the PT100 reference table in Section 8.2.

Symptom	Possible Cause	Corrective Action
Unit does not power on	Blown control fuse	De-energize and lock out all power. Open the enclosure and inspect the two 5 mm × 20 mm, 1 A fuses on the control board. Replace any blown fuse with an identical 1 A type.
RTD channel shows "FAULT" or no reading	Open circuit — broken wire, disconnected terminal, or corroded connector	Inspect the terminal block. Disconnect the sensor and measure resistance at the terminals: a healthy PT100 reads 80–176 Ω across the temperature range in this table. Use of wire ferrules for terminations is advised. Replace cable or sensor if open.
SCADA reads an extreme temperature (approx. 6553 °F) from a faulted channel	Fault value 0xFFFF decoded as unsigned integer	The Modbus master must decode temperature registers as a signed 16-bit integer before dividing by 10. A raw value of 65535 (0xFFFF) is the fault indicator and should be displayed as "FAULT," not as a number.
Temperature reading offset by a constant amount	Sensor type mismatch — PT1000 sensor connected to a PT100-configured channel	Verify the sensor type. The HT-ProMon controller is configured for PT100 sensors. A PT1000 sensor will produce resistance readings approximately 10× higher than expected, resulting in an out-of-range or fault reading.

Heater output turns off, RTD shows fault	RTD sensor failure in the active control mask — controller places heater in safe state	Correct the RTD fault. The heater output will resume automatically once the sensor recovers, without requiring a trip reset.
All three RTD channels fault simultaneously	Common-mode wiring problem — loose supply connection to the sensing circuit	Verify all RTD terminal connections are fully seated. Check that sensor cables are not routed adjacent to high-current load conductors.
No Modbus response from either instance	RS-485 A/B polarity reversed	Swap the A and B conductors at one end of the bus. Refer to the <i>HT-ProMon Series Installation Instructions</i> for correct polarity markings.
No Modbus response	Slave ID mismatch	Verify the Modbus slave ID using the serial diagnostic interface (see Section 7.1). Default: HMI RS485 Instance = ID 1, Remote Monitoring Instance = ID 1.
No Modbus response	Baud rate mismatch	Default baud rate for Instance 1 is 115200 bps; Instance 2 is 9600 bps. Confirm the master's baud rate matches.
No Modbus response	Missing bus termination	Install 120 Ω termination resistors at both physical ends of the RS-485 bus.
Remotely written registers display incorrect register values — temperatures appear too large or too small	Scaling not applied	Temperature register values must be divided by 10. Current (A) values must be divided by 100. Ground fault (mA) values must be divided by 10. Line voltage values must be divided by 100.
Remotely read registers show negative temperature reads as a large positive number	Unsigned integer decode	Decode temperature registers as signed 16-bit integers. Do not interpret the raw register as an unsigned value.

Writing coil 21 to reset trip has no effect	Incorrect coil write value	Write 0xFF00 to coil address 21 using FC05. In Modbus RTU, coil ON is represented by 0xFF00, not 0x0001 or any other value.
Register write appears accepted but value does not change	Value outside valid range for that parameter	Verify the value is within the allowed range (for example: Control RTD Mask must be 1–7; baud rate must be one of 9600, 19200, 38400, 57600, or 115200). Read back the register after writing to confirm.

8.2 PT100 Resistance Reference Table

Use this table to verify sensor condition during field troubleshooting. Measure resistance between the two sensor conductors (not the shield) at the terminal block with the sensor leads disconnected from the controller.

Temperature	Expected Resistance (PT100)
–50 °C (–58 °F)	80.31 Ω
0 °C (32 °F)	100.00 Ω
25 °C (77 °F)	109.73 Ω
100 °C (212 °F)	138.51 Ω
200 °C (392 °F)	175.86 Ω

A reading of 0 Ω or near 0 Ω indicates a short circuit. A reading of infinite resistance (open circuit) indicates a broken conductor or disconnected sensor. Any reading outside the expected range for the process temperature is grounds for sensor replacement.

9. Maintenance

9.1 Routine Inspection

Inspect the HT-ProMon controller at regular intervals appropriate for the installation environment. At minimum, perform the following checks annually or following any event that may have affected the enclosure (physical impact, flooding, sustained high ambient

temperature):

28. **Enclosure integrity:** Verify that the enclosure cover and gasket are undamaged and that the cover fasteners are fully tightened. A compromised gasket reduces the ingress protection rating and may allow moisture or contaminants to reach the control board.
29. **Conduit and cable seals:** Verify that all conduit entries are sealed and that conduit fittings are tight. Check that cable jacketing is not abraded or cut at entry points.
30. **Heat sink clearance (SSR models only):** Verify that the heat sink fins on the exterior of the enclosure are unobstructed and free of debris, dust accumulation, or physical damage. A blocked heat sink will cause the solid-state relay to overheat and may trigger thermal protection.
31. **Terminal connections:** With all power removed and locked out, verify that RTD sensor and communication terminals are fully seated and show no signs of corrosion or heat damage. **Use of wire ferrules for terminations is advised.**

No periodic calibration is required under normal operating conditions. If measurement accuracy is in question, verify sensor resistance at the terminal block against the values in Section 8.2 and replace the sensor if out of specification.

9.2 Fuse Replacement

The control board contains two (2) replaceable 5 mm × 20 mm, 1 A glass fuses that protect the control circuit. These fuses are not field-serviceable under power.

DANGER: Hazardous voltage is present at supply terminals and load terminals even when the heater output is off. De-energize and lock out all power sources — both the supply circuit and any separately powered load circuit — before opening the enclosure.

Procedure:

1. De-energize and lock out all power to the unit following your facility's lockout/tagout procedure.
2. Verify with a voltage tester that the supply terminals are at zero potential.
3. Open the enclosure cover and, where applicable, rotate the swing panel outward to access the control board.
4. Locate the two 5 mm × 20 mm fuses on the control board.
5. Remove each fuse and test for continuity. A blown fuse will show no continuity.
6. Replace any blown fuse with an identical type: 5 mm × 20 mm, 1 A, glass cartridge fuse. Do not substitute a higher-rated fuse.

7. Return the swing panel to the closed position.
8. Close and secure the enclosure cover, ensuring the gasket seats fully around the perimeter.
9. Restore power and verify normal operation.

If the replacement fuse blows immediately or shortly after power is restored, do not continue replacing fuses.

10. Specifications

Electrical Specifications

Parameter	Value
Supply Voltage	85–305 VAC, single-phase (universal input)
RTD Input Type	3× PT100, 3-wire connection
RS-485 Protocol	Modbus RTU, 2-wire + optional ground
Modbus Baud Rates	9600, 19200, 38400, 57600, 115200 bps
Default HMI Modbus Baud	115200
Default Remote Monitoring Baud	9600

Output and Relay Specifications

Parameter	Value
Alarm Relay	Form C (SPDT) dry contact: 6 A @ 250 VAC / 6 A @ 24 VDC
Control Output — EMR Model	Electromechanical relay; max 30 A per output circuit
Control Output — SSR Model	Solid State Relay with integral heat sink; duty-cycle time-proportioning control
Control Fuses	Two (2) replaceable 5 mm × 20 mm, 1 A glass fuses on control board

Wiring and Installation

Parameter	Value	Application
PCB Power Terminal Blocks (GREEN)	14–6 AWG copper wire only	Fixed Power Line / Load
PCB Pluggable Terminal Blocks (Sensor / Telemetry) (BLUE)	28–16 AWG copper wire	RTD(s)/ RS-485/ Dry Alarm Contacts

NOTE: *Use of wire ferrules for terminations is advised.*

Environmental Specifications

Parameter	Value
Operating Temperature Range (Ambient)	-20F – 120 F
Storage Temperature Range	-20F – 120 F
Enclosure Rating (NEMA/IP)	Rating: NEMA 4X Certified/ IP67
Enclosure UL Listing	508A 4/4X Standard
Enclosure Material	UV stabilized Polycarbonate
Mounting Methods	Wall Mount Bracket (if applicable) Pipe Mount Bracket (if applicable)
Enclosure Type	Hinged-door with swing pan; conduit entry preferred from bottom

Physical Specifications

Parameter	Value
Dimensions	
HT-ProMon-EMR	12 in × 8.25 in × 5.125 in (304.8 mm × 209.55 mm × 130.715 mm)
HT-ProMon-SSR	12 in × 8.25 in × 6.365 in (304.8 mm × 209.55 mm × 161.671 mm)
Weight	
HT-ProMon-EMR	4.8 lbs
HT-ProMon-SSR	10.9 lbs

11. RTD Measurement Accuracy and Resolution

The RTD measurement accuracy and resolution specifications stated in this manual represent the capabilities of the controller's internal measurement circuitry under ideal laboratory conditions. Actual system accuracy and thermal resolution in field applications are highly dependent on external variables beyond the manufacturer's control.

Factors such as extension wire resistance, ambient temperature fluctuations, electromagnetic interference (EMI), proper sensor placement, and the quality of thermal insulation will impact real-world performance. Users must ensure proper installation practices, including the use of shielded, twisted-pair wiring and appropriate calibration procedures, to achieve optimal measurement precision.

12. Warranty and Support

For technical support, troubleshooting, or warranty service inquiries, contact Trasor Corp:

Trasor Corp

305 N Hickory Ave.,

Broken Arrow, OK 74012

(800) 324-1551 | WWW.TRASOR.COM

Authorized Components and Service:

Use only Trasor-approved replacement components and accessories. Unauthorized substitutions, modifications, or repairs may void the warranty and compromise safety and compliance.

For component replacement, authorized service, or warranty claims, contact Trasor Corp or an authorized representative.